

BASE24 Release 6.0 Post Version 10

Implementation Guide



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Preface

Document Control

A description of the document revisions is shown below. Any changes of text from one document version to another are indicated with the symbol “|” in the outside margin.

BASE24 Release 6.0 Post Version 10 A(J. Wishart)
– Sep-2011

BASE24 Release 6.0 Post Version 10 B(J. Wishart)
– Jul-2012

BASE24 Release 6.0 Post Version 10 C(J. Wishart)
– Aug-2012

1: Introduction

The ***BASE24 Release 6.0 Post Version 10 Implementation Guide*** is the primary reference document for information needed to upgrade existing BASE24 Release 6.0 Version 10 systems to include any mandates or modifications to BASE24 available Post Version 10.

As no more releases of BASE24 are available after BASE24 Release 6.0 Version 10, customers may need to implement any mandates or modifications on an individual basis. For this reason, this Implementation Guide is in a different format to previous BASE24 Implementation Guides.

This section provides an overview of the planning requirements for BASE24 users wishing to upgrade to mandates or modifications in BASE24 Release 6.0 Post Version 10. It also provides guidance for using this manual.

Also included in this section is a listing of other documents that a BASE24 user should obtain for implementation planning, the general software requirements for BASE24 Release 6.0 Post Version 10, and a glossary of terms used throughout the guide.

Implementation Planning

BASE24 users must develop their own project plans for upgrading their BASE24 systems to Release 6.0 Post Version 10. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the upgrade method to be used.

ACI has provided BASE24 users with the ***BASE24 Release 6.0 Post Version 10 Implementation Guide*** to assist in preparing project plans. This guide includes a list of services available to the BASE24 user and information that existing BASE24 users will need to upgrade to Release 6.0 Post Version 10.

BASE24 Release 6.0 products include the following:

- ☐ BASE24-atm
- ☐ BASE24-pos
- ☐ BASE24-teller
- ☐ BASE24-remote banking
- ☐ BASE24 Interchange Interfaces
- ☐ BASE24 Value-Added Products:
 - BASE24-card
 - BASE24-customer service
 - BASE24-infoBase
 - BASE24-inventory

Who should use this guide?

There are two audiences for this guide: those who plan and manage the upgrade process and those who implement the upgrade to BASE24 Release 6.0 Post Version 10 mandates or modifications. In either case, the reader should be experienced with BASE24 software, HP NSK products, and the BASE24 user's current configuration. This guide is not directed at new BASE24 users who will be installing a BASE24 application for the first time. New users should refer to the formal BASE24 publications listed under the Publications heading in Section 2.

To upgrade to BASE24 Release 6.0 Post Version 10 mandates or modifications, users must be on BASE24 Release 6.0 Version 10. Refer to the **BASE24 Release 6.0 Version 10 Implementation Guide** to assist in planning the tasks necessary to upgrade to BASE24 Release 6.0 Post Version 10.

Using this guide, you can do the following:

- ☐ Determine what ACI services are available to assist in the upgrade to BASE24 Release 6.0 Post Version 10 mandates or modifications.
- ☐ Develop a project plan to upgrade to BASE24 Release 6.0 Post Version 10 mandates or modifications.
- ☐ Determine staffing and training needs.
- ☐ Gain a detailed understanding of the BASE24 Release 6.0 Post Version 10 mandates or modifications.

Software Requirements

HP NSK Software Requirements

ACI recommends that the BASE24 user be on a current HP NSK-supported operating system to obtain maximum performance and support. However, BASE24 Release 6.0 is currently compatible with the HP NSK Guardian D45 Operating System. Customers choosing to implement Format 2 File support are required to be on a minimum Operating System level of D46.

XPNET

ACI requires that the BASE24 user be on an XPNET 3.0 or 3.1 system.

FIX

FIX 2.28 (or newer) is required to apply any new BASE24 fix files produced after 19 October 2006. FIX 2.28, like the previous version of FIX, will allow application of fix files to a specified file name in a concatenated list of fix files. This option can be combined with the FIRSTSCN and LASTSCN options to apply only fixes to the specified file name in a specified SCN range. As of 19 October 2006, FIX 2.28 (or the most current version of FIX) was automatically shipped with each software order. Please replace your existing FIX object with the most current version.

Conversion Planning

BASE24 users must develop their own project plans for upgrading their systems to BASE24 Release 6.0 Post Version 10 mandates or modifications. Project plans should address such issues as staffing and training requirements, host external message changes, device programming enhancements, hardware and software requirements, and the upgrade method to be used.

In general, all changes for BASE24 Release 6.0 Post Version 10 mandates and modifications can be obtained by requesting the latest fixes from ACI and then retrofitting all fixes to the user's BASE24 Release 6.0 Version 10 modules.

Conversion Program Guidelines

Users of previous releases of BASE24 and previous versions of BASE24 Release 6.0 must run file conversion programs in order to migrate to the latest version of Release 6.0. Information about specific conversion programs can be found in subsequent sections of this document under the applicable enhancements. Following are general conversion program guidelines.

The *UC04 subvolumes (e.g., AT60UC04, BA60UC04, PS60UC04) contain the conversion programs for users migrating from BASE24 Release 5.3 to the latest version of BASE24 Release 6.0. The conversion programs on these subvolumes are kept up-to-date with each new version of BASE24 Release 6.0.

The *UC05 subvolumes (e.g., AT60UC05, BA60UC05, PS60UC05) contain the conversion programs for users migrating from Release 6.0 Version 1 to Release 6.0 Version 2. Also residing on these subvolumes are the Transaction Security Services (TSS) conversion programs, which are used to move key data to the new TSS database. The TSS conversion programs are required only if the user is implementing TSS.

The *UC06 subvolumes (e.g., AT60UC06, BA60UC06, PS60UC06) contain the conversion program for users migrating from Release 6.0 Version 2 to Release 6.0 Version 3. No changes are made to these programs other than necessary fixes.

The *UC07 subvolumes (e.g., AT60UC07, BA60UC07, PS60UC07) contain the conversion program for users migrating from Release 6.0 Version 3 to Release 6.0 Version 4. No changes are made to these programs other than necessary fixes.

The *UC08 subvolumes (e.g., AT60UC08, BA60UC08, PS60UC08) contain the conversion program for users migrating from Release 6.0 Version 4 to Release 6.0 Version 5. No changes are made to these programs other than necessary fixes.

The *UC09 subvolumes (e.g., AT60UC09, BA60UC09, PS60UC09) contain the conversion program for users migrating from Release 6.0 Version 5 to Release 6.0 Version 6. No changes are made to these programs other than necessary fixes.

The *UC0A subvolumes (e.g., AT60UC0A, BA60UC0A, PS60UC0A) contain the conversion program for users migrating from Release 6.0 Version 6 to Release 6.0 Version 7. No changes are made to these programs other than necessary fixes.

The *UC0B subvolumes (e.g., AT60UC0B, BA60UC0B, PS60UC0B) contain the conversion program for users migrating from Release 6.0 Version 7 to Release 6.0 Version 8. No changes are made to these programs other than necessary fixes.

The *UC0C subvolumes (e.g., AT60UC0C, BA60UC0C, PS60UC0C) contain the conversion program for users migrating from Release 6.0 Version 8 to Release 6.0 Version 9. No changes are made to these programs other than necessary fixes.

The *UC0D subvolumes (e.g., AT60UC0D, BA60UC0D, PS60UC0D) contain the conversion program for users migrating from Release 6.0 Version 9 to Release 6.0 Version 10. No changes are made to these programs other than necessary fixes.

The *UC0E subvolumes (e.g., AT60UC0E, BA60UC0E, PS60UC0E) would contain any conversion programs for users upgrading from Release 6.0 Version 10 to Release 6.0 Post Version 10 mandates and modifications from 2010. No such conversion programs are needed.

The *UC0F subvolumes (e.g., AT60UC0F, BA60UC0F, PS60UC0F) contain the conversion program for users upgrading from Release 6.0 Version 10 to Release 6.0 Post Version 10 mandates and modifications from 2011. No changes are made to these programs other than necessary fixes.

If a Release 5.3 user wants to convert to the latest Version of Release 6.0, the conversion programs on *UC04 should be run. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run.

If a Release 6.0 Version 1 user wants to convert to Release 6.0 Version 2, the conversion programs on *UC05 should be run.

If a Release 6.0 Version 2 user wants to convert to Release 6.0 Version 3, the conversion programs on *UC06 should be run.

If a Release 6.0 Version 3 user wants to convert to Release 6.0 Version 4, the conversion programs on *UC07 should be run.

If a Release 6.0 Version 4 user wants to convert to Release 6.0 Version 5, the conversion programs on *UC08 should be run.

If a Release 6.0 Version 5 user wants to convert to Release 6.0 Version 6, the conversion programs on *UC09 should be run.

If a Release 6.0 Version 6 user wants to convert to Release 6.0 Version 7, the conversion programs on *UC0A should be run.

If a Release 6.0 Version 7 user wants to convert to Release 6.0 Version 8, the conversion programs on *UC0B should be run.

If a Release 6.0 Version 8 user wants to convert to Release 6.0 Version 9, the conversion programs on *UC0C should be run.

If a Release 6.0 Version 9 user wants to convert to Release 6.0 Version 10, the conversion programs on *UC0D should be run.

If a Release 6.0 Version 10 user wants to upgrade to Release 6.0 Post Version 10 mandates and modifications from 2010, the conversion programs on *UC0E should be run.

If a Release 6.0 Version 10 user wants to upgrade to Release 6.0 Post Version 10 mandates and modifications from 2011, the conversion programs on *UC0F should be run.

If a Release 6.0 user wants to convert from Version 1 to Version Post 10 mandates or modifications from 2011, multiple conversion programs must be run starting with *UC05. If the user is implementing TSS, the TSS-specific conversion programs on *UC05 should also be run. Following this, the user needs to run the conversion programs on *UC06, *UC07, *UC08, *UC09, *UC0A, *UC0B, *UC0C, *UC0D and UC0E. Finally, the user should run the conversion programs on *UC0F.

Other Documents

In addition to this guide, the reader can obtain other BASE24 and NET24-XPNET publications that cover operational and product-specific information required to fully understand and implement the BASE24 Release 6.0 Version Post 10 software. Such manuals include the formal publications mentioned under the Publications heading in Section 2, the event message documentation located on the BA60LOGM and OK60LOGM subvolumes, and the following documents:

BASE24 6.0 Installation Guide (Informal) - BA-AE000-55

A primary reference document for information about the installation of a BASE24 Release 6.0 system from scratch. This document is available on InfoLink.

The BASE24 6.0 Installation Guide is also provided with the software, located on the BA60UD04 subvolume as a file called BASE24. On this subvolume there is also a readme file called AAREAD04, which details how to move the Installation Guide to your hard drive and open it with Microsoft Word.

BASE24 Release 6.0 Interchange Interface Implementation Guide (Informal) - SW-DS0000-60

Describes the function and purpose of ACI's Release 6.0 Switch Interfaces. It explains how customers can implement Release 6.0 Switch Interfaces in their existing networks as well as in a BASE24 Release 6.0 network. It also explains how customers can implement their Fortify.1/Fortify.3 Interfaces in a Release 6.0 network. This document is intended to assist customers in configuring and controlling their Release 6.0 Switch Interchange Interfaces in all BASE24 release networks and their Fortify.1/Fortify.3 Interchange Interfaces in a BASE24 Release 6.0 network. This document is included as a Microsoft Word document and is shipped with the Release 6.0 Interchange Interface software. This document is named NSTLSW60 and is located on the SW60UD04 subvolume.

Note: This guide is to be used with BASE24 releases 4.0, 5.0, 5.1.2, 5.3, and 6.0.

Release 6.0 Data File Impacts Document (Informal)

A primary reference document for information about the data file impacts and the methods that can be used to upgrade to BASE24 Release 6.0 Post Version 10 mandates and modifications. This document is included as a HP NSK TGAL document and is shipped with the BASE24 Release 6.0 Post Version 10 mandates and modifications software. This document is named AAREAD0F and is located on the BA60UC0F subvolume.

HELP24 and HELP24 InfoLink

HELP24 and HELP24 InfoLink are excellent sources of assistance and information. The following access options appear on the support page of the ACI Worldwide Inc. Internet website (<http://www.aciworldwide.com/support>).

- ☐ A current listing of HELP24 phones worldwide.
- ☐ An option to send an e-mail message to HELP24.
- ☐ An option to sign up for HELP24 InfoLink.
- ☐ An option to access HELP24 InfoLink once you have received a user ID and password.

Once you can log on to HELP24 InfoLink, you can do the following:

- ☐ Access the HELP24 database to open a new case.
- ☐ Access installation and fix release procedures online.
- ☐ Access all BASE24, BASE24-eps, and NET24-XPNET documentation online.
- ☐ Search the site for information about BASE24.

The **HELP24 eSupport Users Guide** contains procedures for opening a new case, accessing the database, and accessing documentation on-line. Once you log on to HELP24, the path to the **HELP24 eSupport Users Guide** is Support Center > InfoLink > HELP24 eSupport Users Guide. Once you log on to HELP24, the path to the installation procedures is Support Center > InfoLink > Product Info > BASE24 > BASE24 6.0 Install Info and Guides. The menu displayed gives you three options:

- ☐ Pre-Install Checklists - for BASE24 Release 6.0 and ITS
- ☐ Install Guides - Procedures for installing a software release or version for the first time.
- ☐ Retail Install Guides - Procedures for installing Retail 3.1 or Check Auth 3.2.

Once you log on to HELP24, the path to the Fix Information is Support Center > InfoLink > Fix Information > BASE24

Definition of Terminology

The following list has been defined to assist the reader in understanding the terms referred to within this document.

6.0 Interchange Interfaces

The BASE24 Interchange Interfaces developed in conjunction with BASE24 Release 6.0. These replace the Fortify.3 Interchange Interfaces and can be used with releases 4.0, 5.0, 5.1, 5.1.2, 5.3 and 6.0 of BASE24.

Accountholder

A person for whom a financial account has been established for some purpose. Accountholders may or may not use cards to initiate transactions on their accounts.

BASE24 User

An entity that has licensed BASE24 software products.

Cardholder

A person who has been issued a plastic card for the purpose of initiating transactions.

Consumer

A person who utilizes services offered by a BASE24 user who may or may not require a plastic card.

Conversion Program

A utility within existing code that reads the records from an input file, adds new fields and initializes appropriate fields and then writes the record to an output file. The input file (existing file) and output file (converted/initialized file) are different files. This method (versus the initialization program and online conversion) is required when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.). This method can also be used to initialize existing user fields in a record.

This method requires system down time to convert the file. It also requires an operator to manually rename the file that has been converted, move in the new objects of modules that access the file (e.g., device handler, auth, etc.) and then restart the objects. This method impacts all BASE24 users who use the file being converted even if they are not taking advantage of the enhancement for which the modifications are being made.

Extended Memory Table file (EMT)

A data file produced by the EMT Build utility program. Once the EMT has been built with the EMT Build utility, processes that use the EMT must be warmbooted (or started if not running).

Extended Memory Table Build utility (EMT Build)

The EMT Build utility is a stand-alone program used to build tables in extended memory from information contained in disk files. This utility can build hash tables, radix tables, or custom designed tables. Refer to the ***BASE24-atm Transaction Processing Manual***, ***BASE24-pos Transaction Processing Manual***, or the ***BASE24 ITS Kernel Configuration Manual*** for additional information about this utility.

Online (Inline) Conversion

A software routine within existing code that reads and updates each record in a file to initialize user fields that have been redefined as new fields for use by an enhancement. The file is converted in place (i.e., a new file is not created). The record is read, updated and then rewritten to the same file. This method (versus the conversion program) cannot be used when the record structure changes (e.g., increasing the record length, adding/deleting fields from segments, etc.).

The online conversion code will be executed each time a record is accessed (versus the conversion and initialization programs that are run once). When a record is read, the process will determine if the value of a new field is valid. If the value is not valid, the field will be initialized to the default value. The database is gradually converted as records are read and updated.

This method requires an operator to stop a process, roll in the new object and start processes that use the files that are impacted. This method does not require any additional steps to be performed by an operator (e.g., running conversion/initialization programs, renaming files).

Operator

A person who carries out network operations functions, such as starting and stopping the system, entering ONCF commands, controlling logging, dealing with event messages, performing refreshes and extracts. The term *operator* also can refer to persons using BASE24 screens for such purposes as file maintenance, perusal, etc.

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Standard Information Format

A standard format is used in Sections 3 and beyond to highlight and describe the various changes, conversion requirements, and implementation requirements BASE24 users need to consider when upgrading to BASE24 Release 6.0 Post Version 10 mandates and modifications. This standard format is defined below.

In situations where the following information does not apply, the information is labeled as “Not applicable”

Prerequisites

Describes the software modules that must be licensed in order for this mandate or modification to be implemented.

Software Impacts

Describes the software modules impacted for a given mandate or modification and the nature of the impacts. This information includes a high level list of the modules impacted, how the modules are impacted (i.e., inline changes, hooks, dummy procs, extensions) and which customers are affected (i.e., all, or only those implementing the mandate or modification).

CUSTMACS Changes

Describes CUSTMACS changes required to implement a given mandate or modification. This information includes Make defines that have been added, deleted, or modified.

LCONF Assign Changes

Describes LCONF assign changes required to implement a given mandate or modification. This information includes assigns that have been added, deleted, or modified, along with an explanation of how these assigns can or must be set.

LCONF Param Changes

Describes LCONF param changes required to implement a given mandate or modification. This information includes params that have been added, deleted, or modified, along with an explanation of how these params can or must be set.

Database Configuration

Describes data file changes required to implement a given change or enhancement. This information includes file fields that have been added, deleted, or modified, along with an explanation of how these fields can or must be set. It also identifies file records and record segments that must be added, deleted, or modified.

New Data Files

Identifies new data files that must be created in order to implement a given mandate or modification.

Template Changes

Identifies file templates that must be created or modified in order to implement a given mandate or modification. It also identifies COBNAMES and Device Handler Names (xxxxNAMS) file modifications that must be made.

Device Configuration

Identifies device configuration changes that are required in order to implement a given mandate or modification.

Mega Table (MEGATBL) Changes

Identifies Mega Table changes that are required in order to implement a given mandate or modification.

PI-Table Changes

Identifies Product Indicator (PI) Table changes that are required in order to implement a given mandate or modification.

SEG-Table Changes

Identifies Segment Indicator (SEG) Table changes that are required in order to implement a given mandate or modification.

Security Table (SECTBL) Changes

Identifies Security Table changes that are required in order to implement a given mandate or modification.

PATHCONF Changes

Identifies PATHCONF changes that are required in order to implement a given mandate or modification.

CRT Access Security Changes

Identifies CRT Access security changes that are required in order to implement a given mandate or modification. This includes changes that need to be made to BASE24 operator security records.

Routing Changes

Identifies routing and configuration changes that are required in order to implement a given mandate or modification. This includes required changes to files such as SPROUTE and the BASE24-pos Authorization Selection Table (AST).

Network Configuration

Identifies configuration changes that are required in order to implement a given mandate or modification. This includes NEF changes.

File Conversions

Identifies files that need to be converted in order to implement a given mandate or modification. This implies that a given file could be impacted by multiple mandates and/ or modifications. BASE24 users need to consider all of the impacts to any one specific file before the actual conversion of a file.

Host/Co-Network Impacts

Identifies impacts to a user's host or co-network as a result of a given mandate or modification. This includes changes to the external messages used to communicate with hosts and co-networks as well as changes to refresh and extract formats.

User/Operational Impacts

Identifies potential changes to a user's internal BASE24 operating procedures. This includes changes to operations and file maintenance screens, operator-initiated commands, and reports.

Vendor Impacts

Identifies any issues that need to be addressed with vendors in order to implement a given mandate or modification. This includes hardware, firmware, and software changes that vendors need to make in order for their products to function with the given mandate or modification.

Reference Documents

Lists any outside sources used in the design of a given mandate or modification. BASE24 users can refer to these documents for additional information.

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2: Support Services

This section summarizes the various support services available from ACI. These support services are described as follows:

- ☐ Product consulting and project management
- ☐ Publications
- ☐ Education
- ☐ HELP24
- ☐ ASTech services

Other services, besides the traditional Installation service, such as go-live support, onsite conversion assistance, and software methodology and utility training are available or are being planned. Please contact your ACI Customer Manager for more details of the services ACI can offer.

Product Consulting and Project Management

No formal service programs have been created by ACI for upgrading customers to BASE24 Release 6.0 Post Version 10 mandates and modifications. Customers need to review this entire document and any referenced manuals, to determine what they will need to do to upgrade their BASE24 system.

After reviewing all materials, customers should contact their ACI Customer Manager to discuss any services, such as project management, custom code upgrades, onsite staff, etc., required for upgrading.

Project Planning Guide

The following Project Planning Guide is designed to assist in the successful upgrade of a BASE24 system to BASE24 Release 6.0 Post Version 10 mandates and modifications.

Following this project guide helps ensure the realization of the goals and business objectives of the BASE24 user. Customers requiring further details are encouraged to contact their ACI Customer Manager.

1. Identify Project Team
2. Determine Hardware Requirements
3. Identify Communication Requirements
4. Update HP NSK Software Requirements
5. Upgrade to XPNET 3.0 or 3.1
6. Project Scope Meeting
7. Migration Planning
8. Develop Project Scope Document
9. Define Requirements Definition
10. Develop Project Plan
11. Identify Customer Specific Modification Development
12. Define Certification Requirements
13. Establish Test Facility
14. Install Production Hardware
15. Attend BASE24 Education Classes
16. Develop Internal Procedures
17. Develop System Test Plan
18. Develop Host Software
19. Install BASE24

- 20. Convert Data Files
- 21. Certify Interchange(s)
- 22. Perform System Testing
- 23. Develop Conversion Plan
- 24. Go-Live
- 25. Complete Post Implementation Review
- 26. Turn Over to Customer Management

Publications

The BASE24 Release 6.0 documentation set contains the most current version of each document.

If a BASE24 manual has not been updated for BASE24 Release 6.0, the manual for the prior release appears on the BASE24 Release 6.0 menu with a note identifying the prior release.

The BASE24 documentation set has been updated to include the BASE24 Release 6.0 Post Version 10 mandates and modifications. The updated BASE24 documentation is available in three formats.

- ☐ **CD-ROM.** The CD-ROM contains all PDF files for the manuals in the documentation set.
- ☐ **HELP24 eSupport InfoLink.** All files for the manuals in the BASE24 Release 6.0 documentation set are located on the HELP24 eSupport InfoLink database. The files for individual manuals can be copied from the HELP24 eSupport InfoLink database.

In addition, the HELP24 eSupport InfoLink database contains the latest documentation including additions or revisions created between ZIP files or CD-ROMs.

- ☐ **Software Management Electronic Distribution.** ZIP or ZIP Executable files that contain the same contents as the CD-ROM are available through the ACI Electronic Distribution system (ED).

Documentation for XPNET 3.0 and XPNET 3.1 is available in a separate set titled NET24-XPNET 3.1 that is available in the same formats as the BASE24 documentation.

BASE24 customers can download files from the HELP24 eSupport InfoLink database or order the documentation ZIP files or CD-ROM from their ACI Customer Manager.

Education

BASE24 Technical Education offers a complete set of one to four day modules covering the components of the BASE24 system. BASE24 courses are structured into a modular format that eliminates repetitious material, provides information specific to certain job functions, offers hands-on lab experience, and enhances retention of information taught.

A BASE24 Enhancement course is also available upon request for existing customers upgrading to BASE24 Release 6.0 Post Version 10 mandates and modifications.

The modular format enables BASE24 users to choose modules applicable to their employees' job responsibilities. This provides consolidated training time and offers opportunities for additional employees to receive education.

The course descriptions, outlines and current education schedule can be viewed on the ACI Worldwide web site at www.aciworldwide.com.

HELP24

ACI Worldwide's eSupport site is available for customers to report problems or ask questions about BASE24 Release 6.0 Post Version 10 mandates and modifications. eSupport is an internet web based application that interfaces with the HELP24 Salesforce application.

With ACI eSupport, a customer can:

- ☐ See past and current cases and create new cases.
- ☐ Search KnowledgeBase solutions for support case solutions.
- ☐ See reports for cases by Product and Status.
- ☐ See other relevant technical documents.

An eSupport option, InfoLink, is the link to a wealth of information that can help in day to day business situations. It allows on-line access to ACI Publications, Fix History Information, HELP24 Hotlines and much more. [An eSupport customer ID can be obtained by sending an email to globalhep24support@aciworldwide.com and requesting an eSupport Portal customer ID and password.](mailto:globalhep24support@aciworldwide.com)

As a result of reviewing new troubleshooting methods, HELP24 has acquired a tool called WebEx. This is a state of the art web-based technology that allows ACI to troubleshoot issues in a secure, online environment. Compared to the dial-up environment used in the past, WebEx provides convenient access, a more stable connection and most importantly, it provides a collaborative environment which also allows customers to actively participate in the session by viewing and asking questions about everything that goes on.

To address audit security requirements, WebEx provides the option to record the session so all steps can be documented in a file that can be accessed after the fact if necessary. WebEx has a file transfer option that provides the ability to transfer files from a customer site to ACI, or vice versa. This feature can be used in the event the file(s) are too large to email. [The ACI File Manger tool can also be used to send in files for cases, this tool can be obtained from Electronic Distribution \(ED\).](#)

ASTech Services

ACI offers customer support services through our ASTech Division which allows the selection of a la carte support services to meet individual customer requirements. ASTech, ACI Support and Technical Services, is a service support team which provides the customer with a variety of on-site and remote technical support. ASTech provides customers with supplemental staffing, system performance analysis, review and/or upgrading of previous release customer-specific software development, and detailed operational technical training.

The following are examples of the various services ASTech could undertake to assist during the BASE24 upgrade.

Your ACI Customer Manager can provide you with details of services available in your region that have been tailored to the special needs of different areas of the world.

Customer Software Analysis

Conversion Planning

Upgrade Custom Code

System Test Planning

Go-Live Support

Post Go-Live Support

3: NCR Enhanced AKDS

This section describes the BASE24 implementation requirements associated with the NCR Enhanced AKDS modification.

The implementation requirements are identified using a standard format defined in Section 1.

The BASE24-atm NCR 5XXX Device Handler has been modified to support the NCR Enhanced Remote Key Loading (RKL) scheme for Automated Key Distribution System (AKDS). The existing AKDS bindable extension, AT60AN50.N50AKDS, has been modified to support both NCR's Basic and Enhanced Schemes in the same device handler.

This modification enables customers to perform AKDS downloads from the NCR Device Handler to NCR ATMs configured to support Enhanced Scheme AKDS.

AKDS commands can be performed via NCPCOM, or the Network Control Supervisor (NCS) AFT screen, or via the Device Control Terminal (DCT) AFT screen.

Protocol Details

NCR offers two versions of RKL, called Basic Protocol and Enhanced Protocol.

The Basic Protocol is the RKL first implemented in the EPP, and can be found in all EPPs running firmware version 1.xx and 7.xx. It consists of:

- ☐ Host Public Key signing by NCR, offline. This is a one-time event. The host must have their public key signed so that the EPP can authenticate the key.
- ☐ Host and EPP mutual Authentication. This is the online exchange of certificates between the host and the EPP. The EPP sends its Serial Number and its public key signature to the host. The host sends its public key and signature to the EPP. The host public key is the one provided by NCR in the step above.
- ☐ Host transport of encrypted key to EPP. The host sends an encrypted TDES key certificate to the EPP.

This is the standard RSA protocol for the unilateral transport of a key. The Basic Protocol was found to present PCI security failures. NCR made changes to address these audit gaps and released the Enhanced RKL Protocol as a result.

The Enhanced Protocol adds two additional functions over the basic protocol: Binding, and Explicit Replay Prevention. In order to support these requirements, an additional pair of RSA keys are introduced to the Host key hierarchy. This pair is used to ensure that only public keys belonging to the correct host can be loaded into the EPP. This pair is known as SK-HostRoot and PK-HostRoot. In this scheme, NCR's private key is used to sign the host root key PK-HostRoot as in the Basic scheme and PK-HostRoot is used to sign PK-Host.

Binding means that once the user key is loaded (either the Host Root Public key or the Host Primary Public Key), it may not be overwritten or replaced, unless the user key is first deleted. The deletion command will require authentication.

To prevent replay attacks, the ATM provides a Random Number at the beginning of the sequence to generate the new master Key. When TSS creates the signature of the new master key, it includes the random number in the data used to generate the signature.

The Enhanced Protocol is designed for EPP firmware version 7.xx. Version 7.xx is designed to be fully compliant with the logical security requirements of the new PCI standards. Version 7.xx contains both Basic and the Enhanced protocols so the migration to the Enhanced protocol can be done at the customers pace.

Functional Overview

The changes necessary for the modification affect the existing AKDS commands supported, and also require new commands to be supported. The Enhanced NCR AKDS solution supports the use of six AKDS commands: LOADPUBLIC, LOADMASTER, LOADALLKEYS, LOADPRIPUB, LOADMASTER/COM and DELPUBLIC.

The LOADPUBLIC command now includes an additional step, to load the Host's Root public key and signature to ATM before loading down the Host's Primary Public key and signature.

The LOADMASTER command now requests a random number from the ATM before requesting the HSM to generate a new master key.

The new DELPUBLIC command shuts down the ATM, and then requests from TSS both the key and signature for the key to be deleted – either the Host Root Public key or the Host Primary Public key. These are passed to the ATM which then deletes the appropriate key from the EPP. If the Host Root Public key is deleted, the Host Primary Public Key is also deleted. If just the Host Primary Public key is deleted, the Host Root Public Key remains in the EPP and can be used to load another Host Primary Public Key. The ATM is not brought back into service when the command is completed.

The new LOADPRIPUB command shuts down the ATM and then loads the Host's Primary Public key only to the ATM. The ATM is not brought back into service when the command is completed. This command can be used to load a new Host Primary Public Key after a DELPUBLIC has been used to delete it.

The LOADALLKEYS command must now perform a DELPUBLIC key command (for the Host Root Public Key and the Host Primary Public Key) before it can perform the LOADPUBLIC command.

In order for customers to be able to run a single command to change their master keys (once AKDS has been performed at the ATM at least one time) without deleting the keys each time with the LOADALLKEYS command, a new LOADMASTER/COM command has been created. This command shuts down the ATM, performs the

LOADMASTER command and then loads the PIN key and the MAC Key (if MACing is configured) to the ATM and brings the ATM into service.

In order to support both the Basic and Enhanced schemes, AKDS capabilities will be requested from the ATM before any of the AKDS commands are processed. This will allow the device handler to determine whether the Basic or Enhanced scheme is being supported by the ATM.

The processing of the Basic scheme will continue unchanged.

In addition, support has been added for the new NCR messages that will allow for EPP serial numbers to be longer than 8 bytes. The existing EEKC message used to request a serial number from the ATM, and the EID message sending the serial number to BASE24, currently support only 8 byte serial numbers. A new version of each of these messages will be available as part of this modification. These new messages will be available for communication with the ATM when the ATM is configured for variable length serial numbers, and can be used in both NCR's Basic and Enhanced scheme AKDS.

Customers wishing to run NCR's Apra Edge or Apra Advance software on an ATM from another vendor that supports longer serial numbers (such as Wincor-Nixdorf) will need a CSM to both TSS and the BASE24 NCR 5XXX Device Handler in order to handle the way in which the signed BASE24 public key(s) must be stored and retrieved.

Please note, the Diebold 912 AKDS signature enhancement from BASE24 Release 6.0 Version 8 has not been updated to support the NCR Enhanced AKDS scheme. Customers can continue to use the Diebold 912 AKDS signature enhancement to perform AKDS on Diebold ATMs running NCR software, but only the basic scheme is supported.

Prerequisites

The Transaction Security Services Version 09.2 Service Pack 6 must be installed prior to implementation of this enhancement.

AT-DA031 – NCR 5XXX Device Handler

CEB-SS200 Subset 1 – NCR AKDS package

Software Impacts

All BASE24 customers:

- ☐ Changes to BA60DDL.DDLGTSS
- ☐ Inline changes to BA60ASRC.SECAKDS, BA60SRC.SECUTILS
- ☐ Updated event (log) message manual BA60LOGM.BASECUTL
- ☐ A new subvolume, BA60UC0F, which includes a documentation file and the ATD conversion program.
- ☐ Inline changes to BA60UC05.AAREAD05 and BA60UC04.AAREAD04

All BASE24-atm customers:

- ☐ Inline changes to AT60DDL.DDLFATD and DDLGATD
- ☐ New DDL Documentation files AT60DDLD.ATD0F and ATDG0F
- ☐ Inline changes to AT60AFT.RQATDS, SCR NATD1 and SVATDTS
- ☐ A new subvolume, AT60UC0F, with all of the files required to run the ATD conversion program.
- ☐ Inline changes to AT60UC05.TSSATDS and AT60UC04.CNVATDS

All BASE24-atm NCR 5XXX Device Handler customers:

- ☐ Inline changes to AT60N50.N50G and N50S
- ☐ Inline changes to AT60AN50.N50AKDG and N50AKDS
- ☐ Inline changes to AT60N50.RQN50LS and SCRNN50L
- ☐ Updated event (log) message manual BA60LOGM.AT5070

A complete list of the files affected by this enhancement can be found in the SCN file BA60UD0F.SCNRK L.

CUSTOMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

Not applicable

Database Configuration

ATM Terminal Data (ATD)

The following new field is updated by the device handler during the AKDS message exchange. It indicates whether the ATM supports the new NCR Enhanced scheme or the original NCR Basic scheme, and will control which of the schemes is performed.

Screen Name: ENHANCED SCHEME
Screen: 22
DDL Name: ATDD1-CORE.ENHANCED-SCHEME
Valid Values: "Y" (ENHANCED SCHEME)
 "N" (BASIC SCHEME)
 " " (UNKNOWN)

Default Value: " " – For NCR ATMs this indicates that AKDS has not yet been attempted. It will be the expected value for all other vendor's ATMs.

The following new field is updated by the device handler during the AKDS message exchange. It indicates whether the ATM supports EPP Serial numbers of longer than 8 bytes and will control whether the messages sent to the ATM support the longer length serial numbers.

Screen Name: VARIABLE LGTH SERIAL NUM
Screen: 22
DDL Name: ATDD1-CORE.VAR-LGTH-EPP-SN
Valid Values: "Y" (VARIABLE)
 "N" (FIXED)
 " " (UNKNOWN)

Default Value: " " – For NCR ATMs this indicates that AKDS has not yet been attempted. It will be the expected value for all other vendor's ATMs.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement NCR Enhanced Scheme AKDS, and the minimum firmware version supported is version 7.xx .

Terminal Software

The NCR ATM must be running NCR's Aptra Advance Software, minimum version 3.04, or NCR's Aptra Edge software, minimum version 2.5.

Registry setting

The following registry entry must be set to enabled at the ATM if the NCR Enhanced scheme is to be performed:

HKEY_LOCAL_MACHINE\Software\Classes\WOSA\XFS_ROOT\SERVICE_PROVIDERS\PIN\GENERAL_CONFIGS key:

UseEnhRemoteKeyLoad

0 (default): Not enabled

1: Enabled.

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File ConversionsATM Terminal Data (ATD)

For users wishing to take the NCR Enhanced AKDS modification, the ATD conversion program must be run to convert the ATD records from the Release 6.0 version 10 format to the Release 6.0 Post Version 10 format.

The ATD conversion program on the AT60UC04 subvolume can be used to convert the TDF records from the Release 5.3 format to the ATD/TDF Release 6.0 Post Version 10 format depending on the device type in the record. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVATDS file. This ATD conversion program is required for any BASE24 customer utilizing the ATD/TDF in transaction processing. Users must create the new ATD/TDF files using the BADDLFUP file.

The ATD conversion program on the AT60UC0F subvolume can be used to convert the ATD records from the Release 6.0 version 10 format to Release 6.0 Post Version 10 format. If modifications to the conversion program are necessary, the source code may be modified by editing the CNVATDS file. Running this conversion program is required for any BASE24 customer utilizing the ATD in transaction processing. Users must create the new ATD file using the BADDLFUP file.

For customers wishing to upgrade to the Transaction Security Services (TSS), the ATD conversion program on the AT60UC05 subvolume can be used to create Channel Security Datasource (CSECD) records while updating ATD records with key locators that point to the appropriate CSECD records.

To compile the conversion program, enter at the TACL prompt:

```
$<volume>.makev500.makev501/out $s.#atdcnv.t,name $atdm,nowait/ cnvatdm –  
force –list –listreplace
```

Release 5.3 users should update the CNVATDR obey file on the AT60UC04 subvolume with the appropriate information. Release 6.0 version 10 users should update the CNVATDR obey file on the AT60UC0F subvolume with the appropriate

information. Customers upgrading to use the Transaction Security Services should update the CNVATDR obey file on the AT60UC05 subvolume with the appropriate information. Both a new and old CSECD file must exist before the conversion is run. The new CSECD file must be created using the FUP utility. The old CSECD file will be the output of a previous ATD conversion for Transaction Security Services or will be an empty file the first time an ATD File conversion is run. The CNVATDR should then be obeyed to execute the conversion program.

For more information on file conversions, refer to the AAREAD04 file on the AT60UC04 subvolume for Release 5.3 customers, to the AAREAD0F file on the AT60UC0F subvolume for Release 6.0 version 10 customers and to the AAREAD05 file on the AT60UC05 subvolume for customers upgrading to the Transaction Security Services.

Release 6.0 version 1, release 6.0 version 2, release 6.0 version 3, release 6.0 version 4, release 6.0 version 5, release 6.0 version 6, release 6.0 version 7, release 6.0 version 8, and release 6.0 version 9 customers should refer to the ***Conversion Program Guidelines*** within Section 1 of this document.

Host/Co-Network Impacts

Super Extract Process

Host code changes are required for all BASE24-atm users that use OMF Extract to accept and process information from the ATD record.

User/Operational Impacts

Security Device Configuration in the Transaction Security Services database

For each security device to be used for public key cryptography (PKI) functions, check the Enable PKI field on the Security Device Configuration window of the ACI Desktop Graphical User Interface. If this field is not checked for a security device, it is not used for PKI functions. This enables you to consolidate all PKI functions in just one or a few devices.

Initial Registration and Certification for the BASE24 System

Initial setup procedures must be carried out for Transaction Security Services before an Authentication and Public Key Exchange sequence or a Master Key Download sequence can take place between BASE24 and an ATM.

The EPP must contain a public and private key pair, a signature of the EPP's Public Key, the Certificate Authority's Public Key, and a unique serial number. All of these are created when the EPP is manufactured.

BASE24's Root Public/Private Key pair must be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security Datasource (HSPKD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair, using the key option of "R" for Root.

In addition, BASE24's Primary Public/Private Key pair must also be created. This key pair is created in a hardware security module and stored in the Transaction Security Services database in the Host Public Key Security Datasource (HSPKD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair, using the key option of "P" for Primary. For customers who wish to generate both key pairs in a single command, send the "Key Generate (KEYGEN) Command" using the key option of "B" for both Root and Primary Key pairs. .

The Root Public Key must be sent to the appropriate Certificate Authority to be verified and signed. In this case, the Certificate Authority will be the NCR trusted center. The NCR trusted center must return BASE24's Root Public Key signature, as well as the NCR trusted center's own Public Key. These values must be entered into the Transaction Security Services database.

For customers who are currently performing NCR's basic scheme AKDS at their ATMs, and who wish to continue using their original BASE24 Public/Private Key pair, this original pair will now be treated as the BASE24 Root Public/Private Key pair. The Primary Public/Private Key pair will need to be generated as described above, but the effective date included in the KEYGEN command must be the effective date of the original key pair. The effective date of the original BASE24 Public Private Key pair

can be ascertained by viewing the Host Public Key Perusal screen through the ACI desktop.

For more information about these initial setup procedures, refer to the **BASE24 Transaction Security Services Processing Guide**.

NCR Device Handler – DCT/NCS/NCPCOM Commands

The following commands are supported in the NCR 5XXX Device Handler. Typically they are issued at the DCT screen, but they can also be issued at the BASE24 AFT NCS screen or through NCPCOM.

☐ LOADALLKEYS

This command allows the authentication and Public Key exchange, the download of the Master Key, and a load of the working MAC and/or PIN Key to be processed in a single command.

The processing starts by shutting down the ATM, and continues with the device handler requesting the capabilities from the ATM. This allows the device handler to determine whether the ATM supports the NCR Basic or Enhanced AKDS schemes.

For the ATMs supporting NCR's Enhanced AKDS, the processing continues by deleting the Host's Root Public Key and Primary Public Key from the ATM. This is achieved by requesting the key and signature from TSS for the Host Root Public Key and sending that data to the ATM, which then deletes the keys. (Deleting the Host's Root Public Key from the ATM also causes the Host's Primary Public Key to be deleted). Then the EPP's serial number and signature are retrieved from the ATM and sent to TSS where they are verified and stored. TSS returns the Host's Root Public Key and signature which are passed to the ATM where they are verified and stored. The device handler then requests the Host's Primary Public Key and signature from TSS, and passes them to the ATM where they are verified and stored. The EPP's Public Key and signature are then retrieved from the ATM and passed to TSS where they are verified and stored. At this point the authentication and public key exchange phase has been completed.

The processing continues with the Master Key load by requesting a random number from the ATM. This is passed to TSS which creates a new Master Key and signature of the key (using HSM calls) and then this data is passed to the ATM. The ATM returns Key Verification Values (KVV's) for the new key, which are passed to TSS for verification.

Processing then continues by loading the new MAC Key to the ATM if MACing is supported by the ATM. Then the new PIN Key is loaded to the ATM.

The final step of processing is to bring the ATM back into service.

This command can be performed as often as is required.

In a typical customer environment, this single command can be used to create a new master key at the ATM.

The syntax of this command when entered via NCPCOM is as follows.

DELIVER PROCESS <process name>, "9500LOADALLKEYS <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the BASE24-atm Terminal Data file (ATD).

Note: If AKDS has never been performed at the ATM, the LOADALLKEYS command will fail. See the Error Conditions section below for more details. This is because the first step in the process is to attempt to delete the Host's Root Public Key from the ATM; since it is not present a command reject B22 will be returned from the ATM. In this circumstance, use the LOADPUBLIC command followed by the LOADMASTER/COM command to complete the AKDS processing for the first time. After AKDS completes this first time successfully, LOADALLKEYS can be used for all future AKDS loads.

☐ LOADPUBLIC

This command allows the authentication and public key exchange to be performed.

The processing starts by shutting down the ATM, and continues with the device handler requesting the capabilities from the ATM. This allows the device handler to determine whether the ATM supports the NCR Basic or Enhanced AKDS schemes.

The EPP's serial number and signature are retrieved from the ATM and sent to TSS where they are verified and stored. TSS returns the Host's Root Public Key and signature which are passed to the ATM where they are verified and stored. The device handler then requests the Host's Primary Public Key and signature from TSS, and passes them to the ATM where they are verified and stored. The EPP's Public Key and signature are then retrieved from the ATM and passed to TSS where they are verified and stored. At this point, the authentication and Public Key exchange phase has been completed.

The ATM is not brought back into service at the end of this command. The customer needs to perform a LOADMASTER followed by a load of the MAC and/or PIN Key or a LOADMASTER/COM command before bringing the ATM into service.

A customer not wishing to use the LOADALLKEYS command can issue a

LOADPUBLIC command, followed by a LOADMASTER command and then loads of the MAC and/or PIN Key to the ATM for the first AKDS load (or a LOADMASTER/COM command). Subsequent AKDS loads would need a DELPUBLIC for the Host Root Public Key first, then the LOADPUBLIC, LOADMASTER and the loads of the MAC and/or PIN Keys to the ATM.

The syntax of this command when entered via NCPCOM is as follows.

DELIVER PROCESS <process name>, "9500LOADPUBLIC <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node.
<terminal ID>: The terminal ID defined for the ATM in the Terminal Data file (ATD).

Note: This command cannot be reissued for an ATM until the Host Public Keys have been deleted from the ATM. This can be achieved either by performing a LOADALLKEYS (which will delete the keys, perform the authentication and Public Key exchange phase and then create the new master key all in a single command) or by performing a DELPUBLIC command for the Host Root Public Key. See the Error Conditions section below for more details.

☐ LOADMASTER

This command allows the new Master key to be created and sent to the ATM.

The processing starts by shutting down the ATM, and continues with the device handler requesting the capabilities from the ATM. This allows the device handler to determine whether the ATM supports the NCR Basic or Enhanced AKDS schemes.

The processing continues by requesting a random number from the ATM. This is passed to TSS which creates a new Master Key and signature of the key (using HSM calls) and then this data is passed to the ATM. The ATM returns Key Verification Values (KVV's) for the new key, which are passed to TSS for verification.

The ATM is not brought back into service at the end of this command. The customer needs to perform a load of the MAC and/or PIN Key before bringing the ATM into service.

This command can be performed as often as is required.

The syntax of this command when entered via NCPCOM is as follows.

DELIVER PROCESS <process name>, "9500LOADMASTER <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the Terminal Data file (ATD).

☐ LOADMASTER/COM

This command allows the download of the Master Key, and a load of the working MAC and/or PIN Key to be processed in a single command.

The processing starts by shutting down the ATM, and continues with the device handler requesting the capabilities from the ATM. This allows the device handler to determine whether the ATM supports the NCR Basic or Enhanced AKDS schemes.

The processing continues by requesting a random number from the ATM. This is passed to TSS which creates a new Master Key and signature of the key (using HSM calls) and then this data is passed to the ATM. The ATM returns Key Verification Values (KVV's) for the new key, which are passed to TSS for verification.

Processing then continues by loading the new MAC Key to the ATM if MACing is supported by the ATM. Then the new PIN Key is loaded to the ATM.

The final step of processing is to bring the ATM back into service.

This command can be performed as often as is required.

The syntax of this command when entered via NCPCOM is as follows.

DELIVER PROCESS <process name>, "9500LOADMSTRCOM <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the Terminal Data file (ATD).

Note: This command is provided to allow customers who are uncomfortable with LOADALLKEYS processing (the deletion of the ATMs Host Public Keys each time the command is run) to be able to create a new Master Key and bring the ATM into service with a single command. The first time an AKDS load is performed, either a LOADALLKEYS command must be used, or a LOADPUBLIC command prior to the LOADMASTER/COM.

☐ DELPUBLIC

This command allows the deletion of the Host's Root Public Key and/or the Host's Primary Public Key from the ATM.

The processing starts by shutting down the ATM, and continues with the device handler requesting the capabilities from the ATM. This allows the

device handler to determine whether the ATM supports the NCR Basic or Enhanced AKDS schemes.

For the ATMs supporting NCR's Enhanced AKDS, the processing continues by deleting the Host's Root Public Key and/or Primary Public Key from the ATM. This is achieved by requesting the appropriate Key and signature from TSS (for either the Host Root Public Key or the Host Primary Public Key) and sending that data to the ATM, which then deletes the key. Deleting the Host's Root Public Key from the ATM also causes the Host's Primary Public Key to be deleted.

The ATM is not brought back into service at the end of this command. The customer needs to perform a LOADALLKEYS before bringing the ATM into service.

The syntax of this command when entered via NCPCOM is as follows.

DELIVER PROCESS <process name>, "9500DELPUBLIC <x> <terminal ID>"

<process name>:	The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node.
<x>:	R (Host's Root Public Key) or P (Host's Primary public key)
<terminal ID>:	The terminal ID defined for the ATM in the Terminal Data file (ATD).

Note: If AKDS has never been performed at the ATM, the DELPUBLIC command will fail. This is because it is attempting to delete a key from the ATM that does not yet exist. A command reject B22 will be returned from the ATM. See the Error Conditions section below for more details. Use a LOADALLKEYS command to exchange the Public Keys and create the new Master Key.

□ LOADPRIPUBLIC

This command allows the download of the Host's Primary Public Key to the ATM.

The processing starts by shutting down the ATM, and continues with the device handler requesting the capabilities from the ATM. This allows the device handler to determine whether the ATM supports the NCR Basic or Enhanced AKDS schemes.

The processing continues as the device handler requests the Host's Primary Public Key and signature from TSS, and passes them to the ATM where they are verified and stored.

The ATM is not brought back into service at the end of this command. The customer needs to perform a LOADMASTER command followed by a load of the MAC and/or PIN Key or a LOADMASTER/COM command before bringing the ATM into service.

The syntax of this command when entered via NCPCOM is as follows.

DELIVER PROCESS <process name>, "9500LOADPRIPUB <terminal ID>"

<process name>: The symbolic name assigned to the NCR 5XXX Device Handler process in the XPNET node.

<terminal ID>: The terminal ID defined for the ATM in the Terminal Data file (ATD).

Note: This command cannot be reissued for an ATM until the Host Primary Public Key has been deleted from the ATM. This can be achieved either by performing a LOADALLKEYS (which will delete the keys, perform the authentication and Public Key exchange phase and then create the new master key all in a single command) or by performing a DELPUBLIC command for the Host Primary Public Key. See the Error Conditions section below for more details.

Error Conditions

It is possible to issue the AKDS command in an incorrect order, in which case various error conditions can occur. Listed below are known error scenarios and their solutions.

- ☐ Issuing a LOADALLKEYS command to an ATM that has never received an AKDS command

Error: The LOADALLKEYS command starts by deleting the Host's Public keys from the ATM. If the ATM has never received an AKDS command, then these keys are not present. The delete command will fail at the ATM, and the ATM will send a specific command reject with reject code B22 to BASE24. See **Note 1** at the end of this section. The following EMS events will appear:

```
11-09-16;09:05:19.798 \K9.$B6N      ACI.SKEL-U.1000      504
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      SHUTDOWN was successful for ATM S92B^N50TCP2
11-09-16;09:05:20.212 \K9.$B6N      ACI.SKEL-U.1000      4258
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities request message sent to device
11-09-16;09:05:20.829 \K9.$B6N      ACI.SKEL-U.1000      4259
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities received from device
11-09-16;09:05:20.831 \K9.$B6N      ACI.SKEL-U.1000      4715
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Signature for the Host Root Public Key is being retrieved for device
11-09-16;09:05:21.302 \K9.$B6N      ACI.SKEL-U.1000      4702
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Key and Signature retrieval successful for the particular key
11-09-16;09:05:21.305 \K9.$B6N      ACI.SKEL-U.1000      4707
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Delete key command for Host Root Public Key sent to device
11-09-16;09:05:23.055 \K9.$B6N      ACI.SKEL-U.1000      5422
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2 - Invalid Encryption Key Size
11-09-16;09:05:23.061 \K9.$B6N      ACI.SKEL-U.1000      4256
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, AKDS command rejected. Terminal software or
      protocol may not support AKDS.
11-09-16;09:05:23.257 \K9.$B6N      ACI.SKEL-U.1000      506
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STARTUP was successful for ATM S92B^N50TCP2
11-09-16;09:05:23.279 \K9.$B6N      ACI.SKEL-U.1000      480
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STATE of ATM S92B^N50TCP2 has changed to OPEN, but ATM is faulted.
```

Solution: Use the LOADPUBLIC command followed by the LOADMASTER/COM command to complete the AKDS processing for the first time. After AKDS completes this first time successfully, LOADALLKEYS can be used for all future AKDS loads.

- ❑ Issuing a DELPUBLIC command to an ATM that has never received an AKDS command

Error: The DELPUBLIC command starts by deleting from the ATM the particular Host Public key specified in the command. If the ATM has never received an AKDS command, then these keys are not present. The delete command will fail at the ATM, and the ATM will send a specific command reject with reject code B22 to BASE24. See **Note1** at the end of this section. The following EMS events will appear:

```

11-09-16;09:45:29.445 \K9.$B6N      ACI.SKEL-U.1000      504
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      SHUTDOWN was successful for ATM S92B^N50TCP2
11-09-16;09:45:29.850 \K9.$B6N      ACI.SKEL-U.1000      4258
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities request message sent to device
11-09-16;09:45:30.462 \K9.$B6N      ACI.SKEL-U.1000      4259
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities received from device
11-09-16;09:45:30.465 \K9.$B6N      ACI.SKEL-U.1000      4715
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Signature for the Host Root Public Key is being retrieved for device
11-09-16;09:45:30.956 \K9.$B6N      ACI.SKEL-U.1000      4702
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Key and Signature retrieval successful for the particular key
11-09-16;09:45:30.962 \K9.$B6N      ACI.SKEL-U.1000      4707
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Delete key command for Host Root Public Key sent to device
11-09-16;09:45:32.909 \K9.$B6N      ACI.SKEL-U.1000      5422
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2 - Invalid Encryption Key Size
11-09-16;09:45:32.911 \K9.$B6N      ACI.SKEL-U.1000      4256
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, AKDS command rejected. Terminal software or
      protocol may not support AKDS.
11-09-16;09:45:33.314 \K9.$B6N      ACI.SKEL-U.1000      506
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STARTUP was successful for ATM S92B^N50TCP2
11-09-16;09:45:33.340 \K9.$B6N      ACI.SKEL-U.1000      480
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STATE of ATM S92B^N50TCP2 has changed to OPEN, but ATM is faulted.

```

Solution: Use the LOADALLKEYS command to exchange the Public Keys and create the new Master Key.

- ☐ Issuing a LOADPUBLIC command to an ATM that has Host Public Keys already installed.

Error: Attempting a LOADPUBLIC command when the ATM already has Host Public Keys installed causes the load of the Host's Root Public key to the ATM to fail. The ATM will send a specific command reject with reject code E01 to BASE24. The following EMS events will appear:

```
11-09-16;08:42:42.071 \K9.$B6N      ACI.SKEL-U.1000      467
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STATE of ATM S92B^N50TCP2 has changed to CLOSED
11-09-16;08:42:42.460 \K9.$B6N      ACI.SKEL-U.1000      504
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      SHUTDOWN was successful for ATM S92B^N50TCP2
11-09-16;08:42:43.259 \K9.$B6N      ACI.SKEL-U.1000      4258
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities request message sent to device
11-09-16;08:42:43.875 \K9.$B6N      ACI.SKEL-U.1000      4259
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities received from device
11-09-16;08:42:43.876 \K9.$B6N      ACI.SKEL-U.1000      4205
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP serial number request message sent to device
11-09-16;08:42:44.484 \K9.$B6N      ACI.SKEL-U.1000      4215
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP serial number received from device, verifying signature
11-09-16;08:42:44.839 \K9.$B6N      ACI.SKEL-U.1000      4244
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP serial number signature verified
11-09-16;08:42:45.704 \K9.$B6N      ACI.SKEL-U.1000      5701
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2 - Function not supported by software
11-09-16;08:42:45.707 \K9.$B6N      ACI.SKEL-U.1000      4256
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, AKDS command rejected. Terminal software or
      protocol may not support AKDS.
11-09-16;08:42:46.107 \K9.$B6N      ACI.SKEL-U.1000      506
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STARTUP was successful for ATM S92B^N50TCP2
11-09-16;08:42:46.118 \K9.$B6N      ACI.SKEL-U.1000      480
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STATE of ATM S92B^N50TCP2 has changed to OPEN, but ATM is faulted.
```

Solution: Use the LOADALLKEYS (which will delete the keys, perform the authentication and Public Key exchange phase and then create the new Master Key all in a single command) or perform a DELPUBLIC command for the Host Root Public Key before retrying the LOADPUBLIC command.

- ❑ Issuing a LOADPRIPUB command to an ATM that has the Host Primary Public Key already installed.

Error: Attempting a LOADPRIPUBC command when the ATM already has the Host Primary Public Key installed causes the load of the Host's Primary Public key to the ATM to fail. The ATM will send a specific command reject with reject code E01 to BASE24. The following EMS events will appear:

```

11-09-16;08:56:05.481 \K9.$B6N      ACI.SKEL-U.1000      467
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STATE of ATM S92B^N50TCP2 has changed to CLOSED
11-09-16;08:56:05.809 \K9.$B6N      ACI.SKEL-U.1000      504
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      SHUTDOWN was successful for ATM S92B^N50TCP2
11-09-16;08:56:06.623 \K9.$B6N      ACI.SKEL-U.1000      4258
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities request message sent to device
11-09-16;08:56:07.279 \K9.$B6N      ACI.SKEL-U.1000      4259
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities received from device
11-09-16;08:56:07.283 \K9.$B6N      ACI.SKEL-U.1000      4714
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Host Primary Public Key is being retrieved for device
11-09-16;08:56:07.929 \K9.$B6N      ACI.SKEL-U.1000      4719
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Host Primary Public Key successfully retrieved for device
11-09-16;08:56:07.952 \K9.$B6N      ACI.SKEL-U.1000      4203
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, Host Public Key download message sent to device
11-09-16;08:56:08.497 \K9.$B6N      ACI.SKEL-U.1000      5701
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2 - Function not supported by software
11-09-16;08:56:08.499 \K9.$B6N      ACI.SKEL-U.1000      4256
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, AKDS command rejected. Terminal software or
      protocol may not support AKDS.
11-09-16;08:56:08.902 \K9.$B6N      ACI.SKEL-U.1000      506
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STARTUP was successful for ATM S92B^N50TCP2
11-09-16;08:56:08.930 \K9.$B6N      ACI.SKEL-U.1000      480
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STATE of ATM S92B^N50TCP2 has changed to OPEN, but ATM is faulted.

```

Solution: Use the LOADALLKEYS (which will delete all the Host's Public keys, perform the authentication and Public Key exchange phase and then create the new Master Key all in a single command) or perform a DELPUBLIC command for the Host Primary Public Key before retrying the LOADPRIPUB command.

- ☐ Using the standard NCR fixed length serial number compatible messages with an ATM that has a serial number greater than 8 bytes long.

Error: Running NCR's older software didn't support the additional variable length serial number field in the capabilities message that is sent to the ATM. Running the standard NCR basic scheme on a ATM with a serial number greater than 8 bytes in length, caused an error condition. Once the LOADALLKEYS or LOADPUBLIC commands were attempted, the ATM would send a specific command reject with reject code E06 to BASE24, to reject the Serial Number request Extended Encryption Key Change (EEKC) message. The following EMS events would appear:

```
11-09-16;08:42:42.071 \K9.$B6N      ACI.SKEL-U.1000      467
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STATE of ATM S92B^N50TCP2 has changed to CLOSED
11-09-16;08:42:42.460 \K9.$B6N      ACI.SKEL-U.1000      504
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      SHUTDOWN was successful for ATM S92B^N50TCP2
11-09-16;08:42:43.259 \K9.$B6N      ACI.SKEL-U.1000      4258
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities request message sent to device
11-09-16;08:42:43.875 \K9.$B6N      ACI.SKEL-U.1000      4259
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP capabilities received from device
11-09-16;08:42:43.876 \K9.$B6N      ACI.SKEL-U.1000      4205
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, EPP serial number request message sent to device
11-09-16;08:42:45.704 \K9.$B6N      ACI.SKEL-U.1000      5701
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2 - Data incompatible with response msg
11-09-16;08:42:45.707 \K9.$B6N      ACI.SKEL-U.1000      4256
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      ATM S92B^N50TCP2, AKDS command rejected. Terminal software or
      protocol may not support AKDS.
11-09-16;08:42:46.107 \K9.$B6N      ACI.SKEL-U.1000      506
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STARTUP was successful for ATM S92B^N50TCP2
11-09-16;08:42:46.118 \K9.$B6N      ACI.SKEL-U.1000      480
      FROM: P92B^N50TCP      RE: S92B^N50TCP2
      STATE of ATM S92B^N50TCP2 has changed to OPEN, but ATM is faulted.
```

Solution: Use a newer version of the NCR software that does support the additional variable length serial number field in the capabilities message that is sent to the ATM. This allows the device handler to send the newer EEKC messages that handle variable length serial numbers, to the ATM.

Note 1

In the situation where the ATM has no Host Public Keys installed, (either because AKDS has not yet been attempted, or because a Delete keys command has been performed) and either a LOADPUBLIC or a DELPUBLIC command is attempted, the next valid AKDS command that is performed at the ATM will cause the ATM to disconnect once it receives the capabilities request.

This issue has been reported to NCR, who are currently investigating. A fix for this issue is expected from NCR in the near future. Please contact NCR for further information.

If this situation occurs, once the ATM reconnects, a LOADPUBLIC keys command can be successfully performed.

Supported Keys

Only the master key (A-key) can be downloaded to an ATM using the AKDS enhancement. AKDS does not support the downloading of the working PIN and/or MAC keys. The DES keys (e.g. PIN encryption key, MAC key) will continue to be encrypted under the master key and downloaded to the ATM as per normal procedures prior to this modification.

Performance Considerations

If the NCR Enhanced AKDS modification is implemented, the additional HSM commands may have an impact on system performance.

Event Messages

New event (log) messages were created as a result of this enhancement. Operators must be trained to recognize the new event messages and perform the appropriate remedy for a given message. Refer to the BA60LOGM.AT5070 and BA60LOGM.BASECUTL event message documentation files.

Vendor Impacts

Compatible Devices and Software

The following device platforms and software versions are compatible with BASE24's NCR Enhanced AKDS.

- An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the RSA scheme.
- Thales e-Security version 6.02 or greater firmware, and the optional DSP plug-in (for RSA functions) provides public key cryptography commands using the RSA scheme.

Encrypting PIN Pad

An Encrypting PIN Pad (EPP) must be installed on every ATM that is to implement AKDS. An EPP is a tamper-proof ATM component comprised of a PIN pad, a security module to perform cryptographic processing, and the cryptographic keys. All of the ATM's cryptographic functions are performed by the EPP. The minimum firmware version required on every ATM that is to implement NCR Enhanced Scheme AKDS is version 7.xx.

Reference Documents

Atalla: *Atalla Key Block Banking Command Reference Manual* (524130-002, July 2002)

Thales: *Host Security Module RG7000 Programmer's Manual* (1270A514 Issue 4, November 2001)

NCR: *Functional Specification for EPP Remote Key Management* (445-0653348 rev F, August 2007)

NCR: *Apra Advance NDC Reference manual* (B006-6180-M000, October 2010)

NCR: *NDC Message Interface Specification* (445-0705907 rev M, November 2010)

4: BASE24 LIS5 and DH Update for Additional Telcos

This section describes the BASE24 implementation requirements associated with the BASE24 LIS5 and DH Update for Additional Telcos modification.

The implementation requirements are identified using a standard format defined in Section 1.

The BASE24-atm NCR 5XXX Device Handler, the BASE24-atm Diebold 10XX/478X Device Handler, the BASE24-atm Triton Device Handler and the BASE24 LINK Interchange Interface have been modified to support additional Telcos.

For the LINK Interface, field 123 Authorization Data contains one or more sub-fields that are used for data related to the transaction. Each sub-field is assigned a unique ID (or 'Tag') that can be used to identify the information the field contains.

BASE24 currently supports 9 Telcos in Field 123 Authorization Data, Sub-field 002, however the 2011 LIS5 Specification has 11 Telcos listed which now need to be supported.

Sub-field 002	Product ID
Tag	002
Length	4
Format	n 4
Presence	Must be present, if applicable to the transaction.
Description	A value that identifies the specific product.

The following values are currently defined for MPTU transaction types:

30nn' : Mobile Phone Top Ups
'32nn' : MONILINK Mobile Phone Top Ups
'34nn' : Internet Banking Top Ups

Where 'nn' identifies the Mobile Telecommunications Provider as follows:

'02' : O2 Mobile
'03' : Tesco Mobile
'04' : Orange
'05' : T-Mobile
'06' : Virgin Mobile
'08' : 3 Mobile
'09' : Vodafone
'10' : Asda Mobile
'11' : Talk Mobile
'12' : Lyca Mobile
'13' : Lebara Mobile

A transaction is acquired by a BASE24 ATM and the consumer selects the Mobile Network Operator (Service Provider) on the ATM screen (NCR, Triton and Diebold).

Upon selection of the Mobile Network Operator, the appropriate Device Handler populates the Track Two data. The Track Two definition now needs to support a two digit product id to allow for the additional telcos.

Functional Overview

The changes to the BASE24-atm NCR 5XXX Device Handler, the BASE24-atm Diebold 10XX/478X Device Handler and the BASE24-atm Triton Device Handler are in the Mobile Top-Up bindable extensions to those device handlers.

The Device Handlers pass the details of the Mobile Network Operator to the LINK Interface by use of the Pre-Pay Top-Up Token (token ID 'BI'). This token definition is therefore changing to allow for a two digit product id.

The Device Handlers use the contents of the LCONF MOBILE-OPERATOR-ID-VTxx Params to build the list of Mobile Network Operators. The values in these LCONF Params will also now change to include two digit product IDs.

The current structure of LCONF params, tables and screens used for Mobile Network Operator information within the NCR and Diebold Device Handlers will allow up to 30 entries. Current LIS5 documentation indicates 13 values for Mobile Network Operator are supported. It is assumed that the 30 entries currently supported in the NCR and Diebold code is sufficient and will not be expanded.

The predefined screens for Triton ATMs and the download request response message to Triton ATMs are not able to handle additional Mobile Network Operator entries returned in FIID 'xq'. The current limit is 8. The device handler will be updated to use a two digit product ID, but a limit of 8 instance of Mobile Network Operator will continue as it is a limitation of the Triton Device.

Prerequisites

MC-PP004 – NCR 5XXX Mobile Top-up Device Handler extension

MC-PP013 – Diebold 10XX/478X Mobile Top-up Device Handler extension

MC-PP015 – Triton Mobile Top-up Device Handler extension

MC-PA005 – Mobile Top-up ATM LINK Acquirer Package

SW-DS969 – Switch LINK ISO Interface

Software Impacts

All BASE24 customers:

- ☐ Changes to BA60DDL.DDLBATKN
- ☐ Changes to BA60MISC.LCONFAT and BAMISCTD

All BASE24-atm NCR 5XXX Device Handler Mobile Top-Up customers:

- ☐ Inline changes to AT60TN50.N50MTUG
- ☐ Inline changes to AT60TN50.N50MTUS

All BASE24-atm Diebold 10XX/478X Device Handler Mobile Top-Up customers:

- ☐ Inline changes to AT60T100.C100MTUG
- ☐ Inline changes to AT60T100.C100MTUS

All BASE24-atm Triton Device Handler Mobile Top-Up customers:

- ☐ Inline changes to AT60TTRT.TRITMTUG
- ☐ Inline changes to AT60TTRT.TRITMTUS
- ☐ Updated Triton Manual documentation AT60TRIT.TRITMAN

All BASE24-atm LINK Interchange Interface customers:

- ☐ Inline changes to SW60LINK.BALINKG
- ☐ Inline changes to SW60LINK.ATLINKS
- ☐ Updated LINK Interface Manual documentation SW60LINK.LINKMNWD
- ☐ Updated LINK Update documentation SW60LINK.LINKUPDT

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

The value in the existing MOBILE-OPERATOR-ID-VTxx parameters (used by the Device Handlers) needs to be changed, to include a two digit product ID instead of the original single digit product ID.

This param specifies a mobile operator six digit IIN, a two digit product ID and the mobile operator name (up to 20 characters).

For example, the existing param MOBILE-OPERATOR-ID-VT02 with a value of:

1234569VODAFONE

would need to be updated to a value of:

12345609VODAFONE

Database Configuration

Not applicable

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Link Interchange

Field 123 sent to Link now contains a 2 byte product ID rather than the single byte product ID that was previously sent. The LINK interface will need to expect this difference.

User/Operational Impacts

Performance Considerations

Not applicable

Event Messages

No new event (log) messages were created as a result of this enhancement.

Vendor Impacts

Not applicable

Reference Documents

Vocalink: *What's New for LINK 2011 Release v1.1.pdf* (LIS5 (2011) V1.0, March 30 2011)

Vocalink: *LIS5 Switch Service Interchange Standard V1.0* (LIS5 2011)

5: BASE24 AMEX GNS iCSC Support

This section describes the BASE24 implementation requirements associated with the BASE24 AMEX GNS iCSC Support modification.

The implementation requirements are identified using a standard format defined in Section 1.

The BASE24-atm Authorization, the BASE24-pos Router Authorization and the card verification security utilities have been modified to support BASE24 AMEX GNS iCSC Support.

The American Express Hardware Security Module (HSM) Vendor Documentation Functionality Overview and personalization bureaus used by American Express Global Network Services (AEGNS) and its participants have received specifications for the Card Security Code (CSC) Version 2.0 algorithm and must support the functionality for iCSC.

Issuers must encode the iCSC instead of the 5CSC on the magnetic stripe equivalent of Chip Cards. iCSC enables issuers to identify fraudulent use of chip data in magnetic stripe transactions and separates the card data stored on the chip from the data stored on the magnetic stripe. This prevents the use of data 'cloned' from a chip to produce a fraudulent magnetic stripe. For new chip card issuers that will begin issuing chip cards after October 31, 2011, for existing chip card issuers issuing new and replacement chip cards after October 31, 2012, and for all chip cards by April 30, 2016, issuers must encode the iCSC value instead of the 5CSC on the magnetic stripe equivalent of AEIPS Chip Cards.

BASE24 is able to verify cards that were issued with card verification values generated using the Card Security Code (CSC) Version 1.0 algorithm or Card Security Code (CSC) Version 1.0 algorithm. The Version 1.0 algorithm does not use service code values in the verification logic. The Version 2.0 algorithm does use service code values in the verification logic.

Service Code Values

The service code values that are used for the CSC Version 2 algorithm are as follows:

'999' - this is the new service code that is used to generate/validate iCSC read from the track data equivalent in a chip read transaction. Users can configure and send this value via EMV-specific fields that were added to the CPF for BASE24 Release 6 version 5 for the 'Visa iCVV Support' enhancement.

Details are included in the BASE24 Release 6.0 Version 5 Implementation Guide under the heading 'Visa iCVV Support'.

'702' - this is the new service code value that is used to generate/validate the iCSC read from the track data equivalent in a contactless chip read transaction. The BASE24 AEGN Interface does not currently support the contactless entry pt-srv-entry-mde values. Contactless support was not added as part of this project. Modifications to POS Router Auth set the service code to '702' for AMEX transactions with a pt-srv-entry-mde value '7' (Integrated circuit card contactless). If contactless support is added to the AMEX GNS Interface in the future, card verification support for the transactions will already exist in Router Auth.

'201' - this is the service code value that is used to generate/validate 5CSC from the magnetic stripe in swiped transactions or chip fallback transactions. Modifications to ATM AUTH and POS Router Auth set the service code to '201' for AMEX transactions to allow 5CSC to be validated.

'000' - this is the standard service code used for manual card verification. TSS will use it as a hardcoded value for validating 3CSC/4CSC.

TSS Changes

All card verification for AMEX cards is currently required to be performed via TSS. Transaction Security Services (TSS) 09.2 Service Pack 9 has been enhanced to support the CSC Version 2.0 algorithm.

The TSS Message 01 is being enhanced as follows:

Card Verify Request Message:

- ☐ Version field - existing field, offset 53 - was enhanced to support new value "02".

Card Verify Response Message:

- ☐ Version field - existing field, offset 4 - was enhanced to support new value "02".
- ☐ Amex CSC Algo - new defined alphanumeric field - offset 9, 1 byte. Possible values: '1' - CSC Version 1; '2' - CSC Version 2. This field is present in the response message only in the case Version is '02' and "Algo" is field is '1' (AMEX) and it indicates which of AMEX CSC Version ('1' or '2') has been used by card verification process.

BASE24 was modified to account for the changes in the card verification message to TSS. A new value for version ID ('02') was introduced for the TSS card verification messages.

Support for a new AMEX_CSC_ALGO environment variable was added for TSS to allow users to configure which AMEX CSC Card Verification Algorithms to execute, and in what order. The possible values are as follows.

- ☐ AMEX_CSC_ALGO is "1" - perform card verification by using AMEX CSC Algorithm Version 1 (this is also the default value).

- ☐ AMEX_CSC_ALGO is "2" - perform card verification by using AMEX CSC Algorithm Version 2.
- ☐ AMEX_CSC_ALGO is "3" - perform card verification by first using AMEX CSC Algorithm Version 1 and if that fails use AMEX CSC Algorithm Version 2.
- ☐ AMEX_CSC_ALGO is "4" - perform card verification by first using AMEX CSC Algorithm Version 2 and if that fails use AMEX CSC Algorithm Version 1.

Assumptions

- ☐ For the newly issued contact chip cards, CSC Version 2.0 algorithm must be used. For these cards, Service code '999' will be used for iCSC generation. Service code '201' will be used for 5CSC generation. Service code '000' will be used for 4CSC and 3CSC generation.
- ☐ For the newly issued contactless chip cards, CSC Version 2.0 algorithm must be used for iCSC generation. For these cards, Service code '702' will be used for iCSC generation.
- ☐ For the already issued contactless chip cards, CSC Version 1.0 algorithm must be used for 5CSC, 4CSC, and 3CSC validation.
- ☐ Issuers using the CSC Version 2.0 to calculate the iCSC may also use it to generate 5CSC, 4CSC and 3CSC values
- ☐ For magnetic stripe only cards, issuers may choose to use either the CSC algorithm Version 1.0 or CSC algorithm Version 2.0 to generate 5CSC, 4CSC, and 3CSC values.
- ☐ Either an iCSC, 3CSC, 4CSC, or 5CSC value will be present in a transaction.
- ☐ An iCSC value may exist for ICC transactions.
- ☐ The 5CSC/4CSC/3CSC value may exist for magnetic swiped transactions.
- ☐ An ICC transaction from AMEX GNS will never have a 5CSC/4CSC/3CSC in DE53, and therefore an ICC transaction will never have DE53 present.

Prerequisites

BA-AE000 – BASE24 generic application

PS-AE010 or PS-AE-040 or PS-AE050 or PS-AE060 – a POS authorization group

AT-AE010 or AT-AE020 or AT-AE030 or AT-AE0400 – an ATM authorization group

PS-CC986 – POS application EMV extension

SW-DS254 – AMEX GNS Interface ATM and POS

TSS 09.2 Service pack 9 or a later version of TSS is required to support this modification.

Software Impacts

All BASE24 customers:

- ☐ Changes to BA60DDL.DDLGTSS
- ☐ Changes to BA60MISC.LCONFBA and BAMISCTD
- ☐ Changes to BA60SRC.CVUTILS and SECUTILS
- ☐ Updated log message documentation BA60LOGM.ATAUTH and PSROUTER

All BASE24-atm customers:

- ☐ Changes to AT60AUTH.AUTHG and AUTHLIBS

All BASE24-pos customers:

- ☐ Changes to PS60RTAU.RTAUG, ROUTERS and AUTHLIBS

All BASE24-pos EMV customers:

- ☐ Changes to PS60IRTA.RTAUEMVS

All BASE24-atm AMEX GNS Interchange Interface customers:

- ☐ Updated AEGN Interface Manual documentation SW60AEGN.AEGNMAN
- ☐ Updated AEGN Update documentation SW60AEGN.AEGNUPDT

CUSTMACS Changes

Not applicable

LCONF Assign Changes

Not applicable

LCONF Param Changes

This section describes the param record that must be added to the LCONF to support these changes. Details of the new parameter are provided in the LCONFBA file located on the BA60MISC subvolume.

AMEX-CSC-ALGO-VER

This param, used by the BASE24-pos Router/Authorization process and the BASE24-atm Authorization process, allows users to configure which version of the TSS card verify request and response messages will be used to communicate card verification data to TSS when verifying AMEX cards. Valid values:

- 1 = Use version 1 of the TSS card verify request and response messages
- 2 = Use version 2 of the TSS card verify request and response messages.

The default value is 1.

Database Configuration

Service code '999' is required to be passed to the card verification logic in TSS in order to validate the new AMEX iCSC values. Fields to allow users to configure a service code value for validating iCVV values were added to the CPF Screen 11 as part of BASE24 Release 6.0 Version 5 for the 'Visa iCVV Support' enhancement. These fields serve the same purpose as is required for configuring AMEX iCSC values and so will be set to '999' to configure service codes to verify iCSC values for this project.

Details are included in the BASE24 Release 6.0 Version 5 Implementation Guide under the heading 'Visa iCVV Support'.

New Data Files

Not applicable

Template Changes

Not applicable

Device Configuration

Not applicable

Mega Table (MEGATBL) Changes

Not applicable

PI-Table Changes

Not applicable

SEG-Table Changes

Not applicable

Security Table (SECTBL) Changes

Not applicable

PATHCONF Changes

Not applicable

CRT Access Security Changes

Not applicable

Routing Changes

Not applicable

Network Configuration

Not applicable

File Conversions

Not applicable

Host/Co-Network Impacts

Not applicable

User/Operational Impacts

Performance Considerations

Not applicable

Event Messages

Users should be aware that additional event messages will be generated as a result of implementing this enhancement.

Vendor Impacts

NSP Atalla Key Block Firmware Version 1.30 or above is required to be in place for customers using Atalla AKB HSMs for AMEX Card Security Code (CSC) Version 2.0 algorithm.

Thales PayShield 9000 version 1.3 or above is required to be in place for customers using Thales HSMs for AMEX Card Security Code (CSC) Version 2.0 algorithm.

Reference Documents

AMEX GNS: Policy and Technical Release Announcement (Oct 2011 for March 2012 Implementation)

AMEX GNS: Network Specification October 2011

Thales: Host Command Reference Manual, Payshield 9000 version 1.3d (January 2012)

Atalla: NSP Atalla Key Block Firmware Version 1.30 (July 2011)